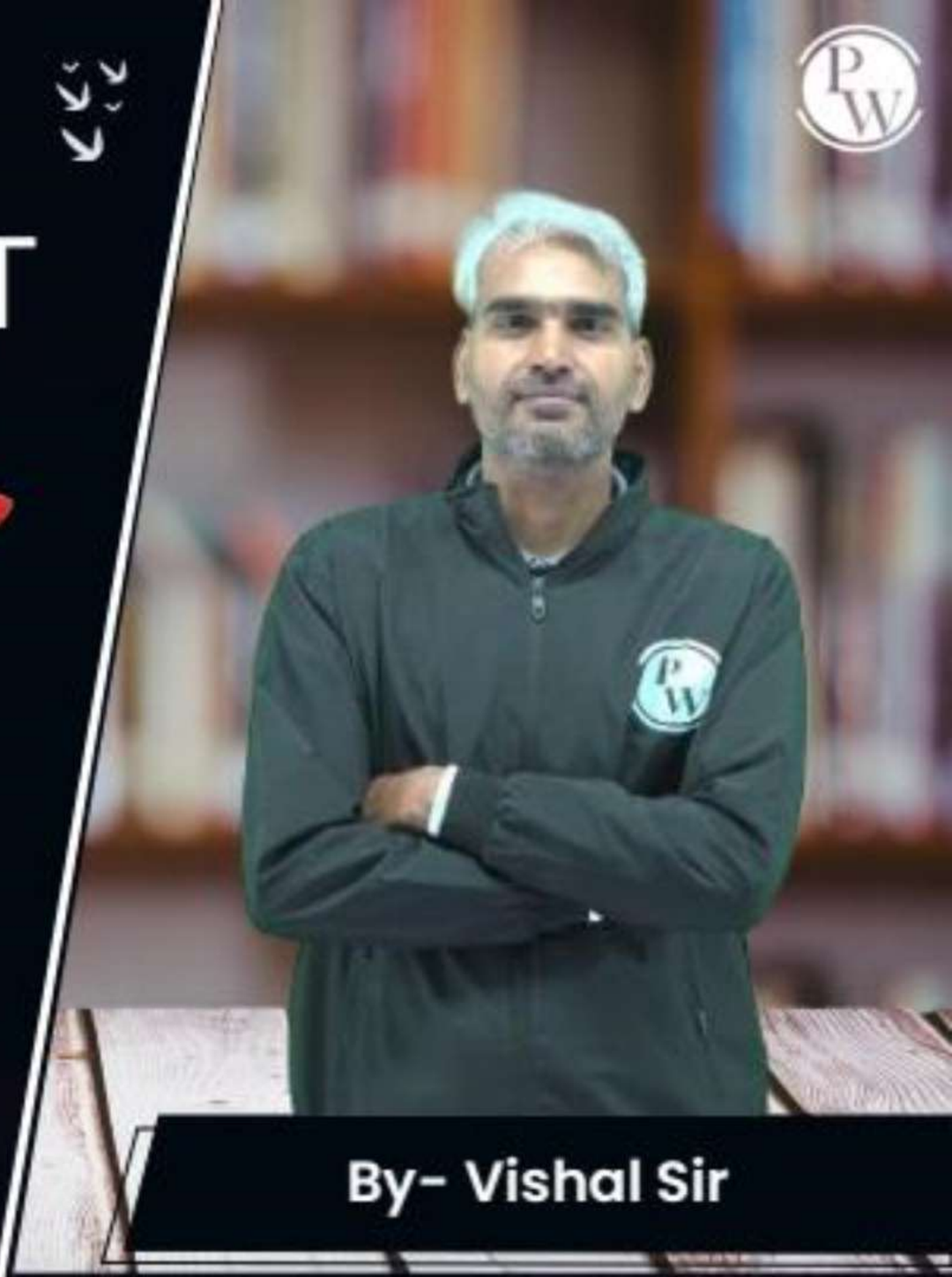
Computer Science & IT



Compiler Design

Introduction and Lexical Analysis

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



Topic — Phases of Compiler

Topic — Translator

Slide

Topics to be Covered

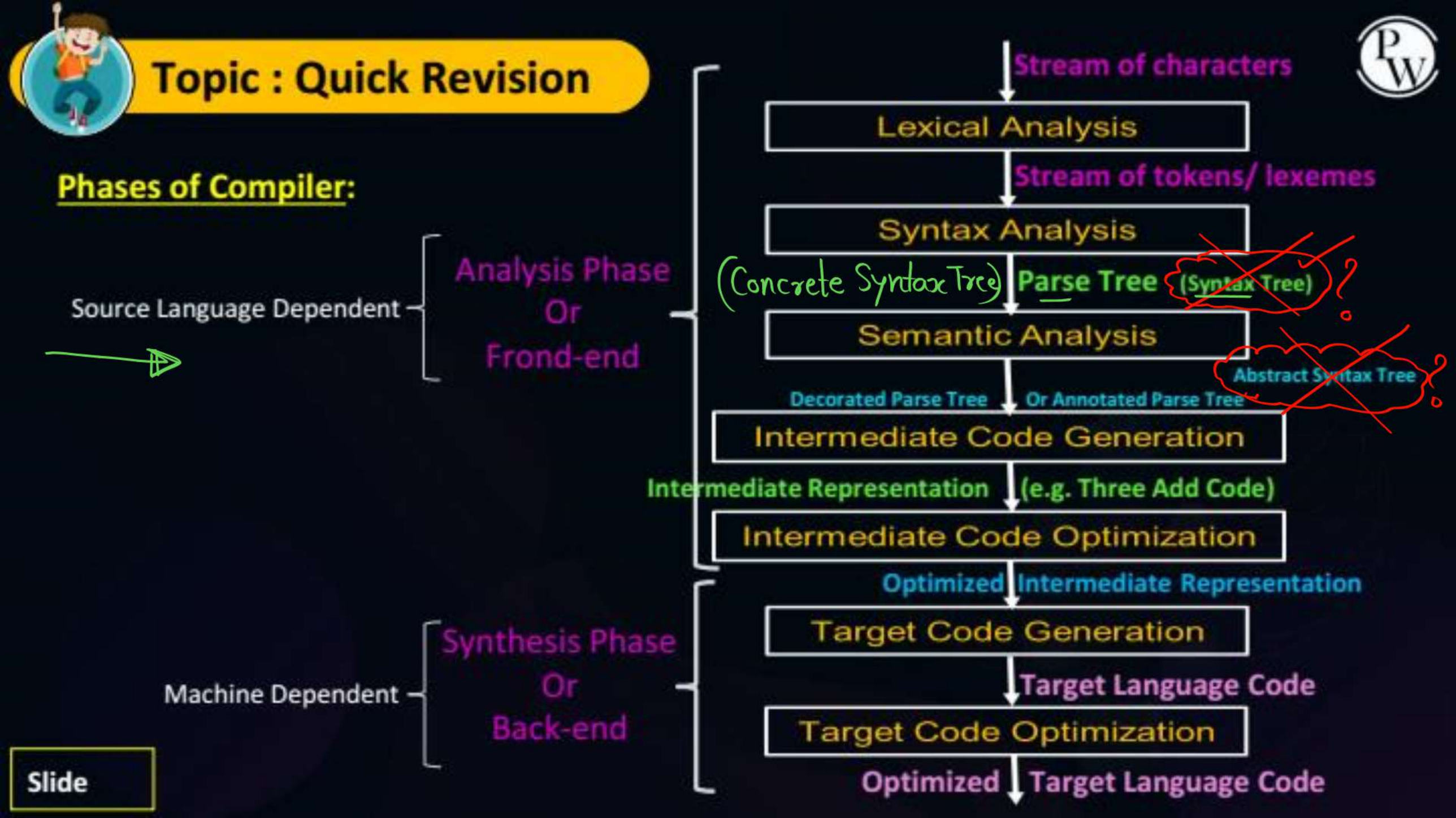


- Topic — Quick Revision
- Topic — Symbol Table & Error Handler
- Topic — Translation example
- Topic — Lexical Analyzer

Slide

Topic : Quick Revision

Phases of Compiler:



Slide



Topic : Quick Revision

Translator:

Compiler:

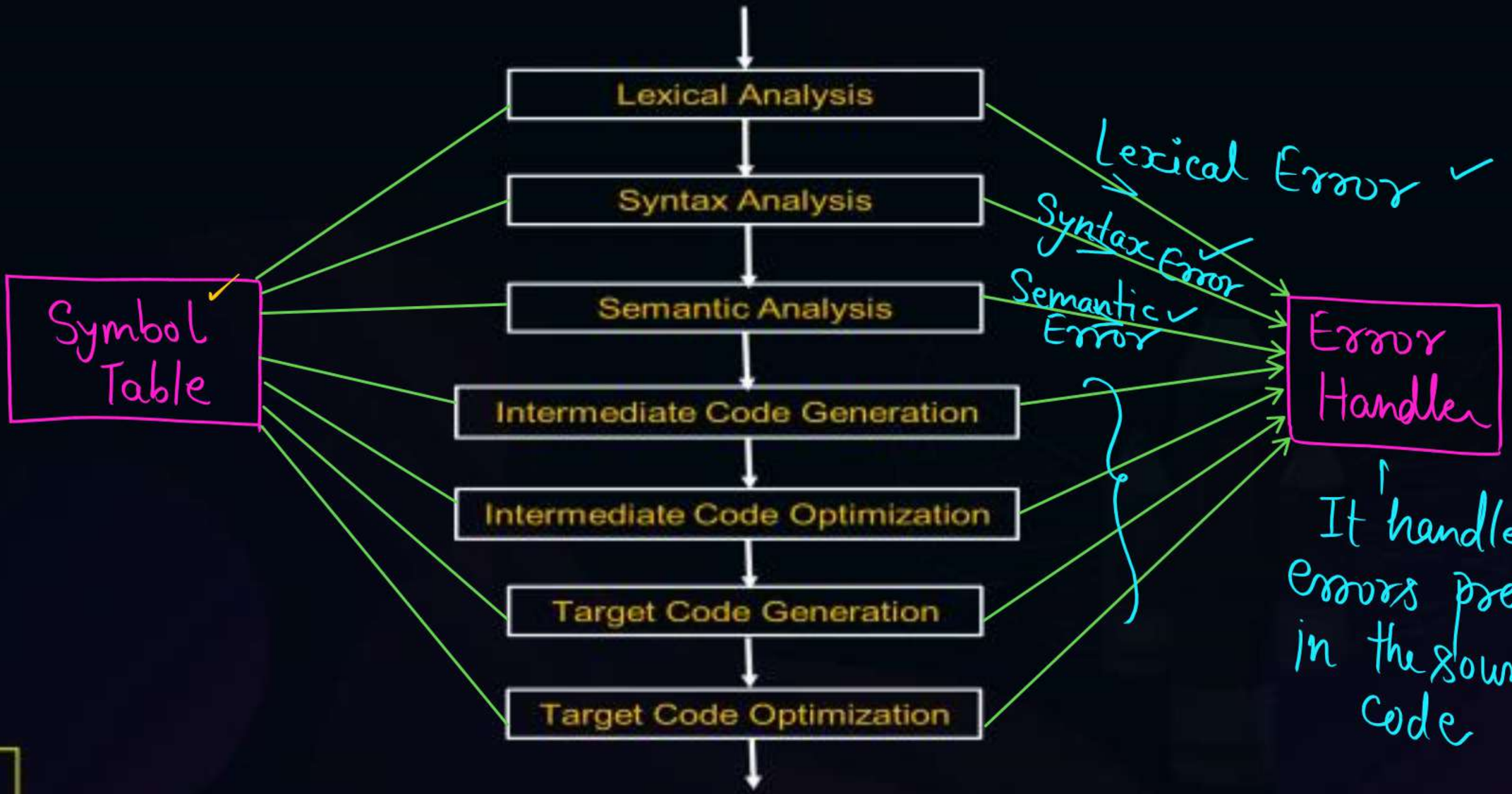
1. A compiler translates complete code from a high-level programming language into machine code before the program runs.
2. Program needs to be compiled only once before execution.
3. Compiled code runs faster (don't need to analyze).
4. Compiler displays all the errors in the source code after analyzing complete code. if code has any error, then it will not compile.

Interpreter:

1. An interpreter translates code written in a high-level programming language into machine code line-by-line as the code runs.
2. Program needs to be interpreted every time it is executed.
3. Interpreted code runs slower (analysis of source code is needed).
4. If there is an error in the statement, then the interpreter terminates its translating process at that statement itself and displays an error message.



Topic : Symbol table & Error handler



Slide



Topic : Symbol table & Error handler

* **Symbol Table :** Symbol table is a data structure used to store the information about all identifiers (variables, functions, constants etc) along with their attribute values

→ * Intially entries are created in the symbol table by lexical analyzer, and all other phases will use and update the information present in the symbol table as needed

Slide



Topic : Symbol table & Error handler

Error Handler : Error handler is a module that is used to handle all the errors present in the given source code. In any phase if any error is observed then it is reported to the error handler and that phase continue with the remaining code.

At the end of Compilation the error handler will display all the reported errors.

Slide



Topic : Translation example

Consider the following grammar

Start symbol

$S \longrightarrow id = E$

$E \longrightarrow E+T/T$

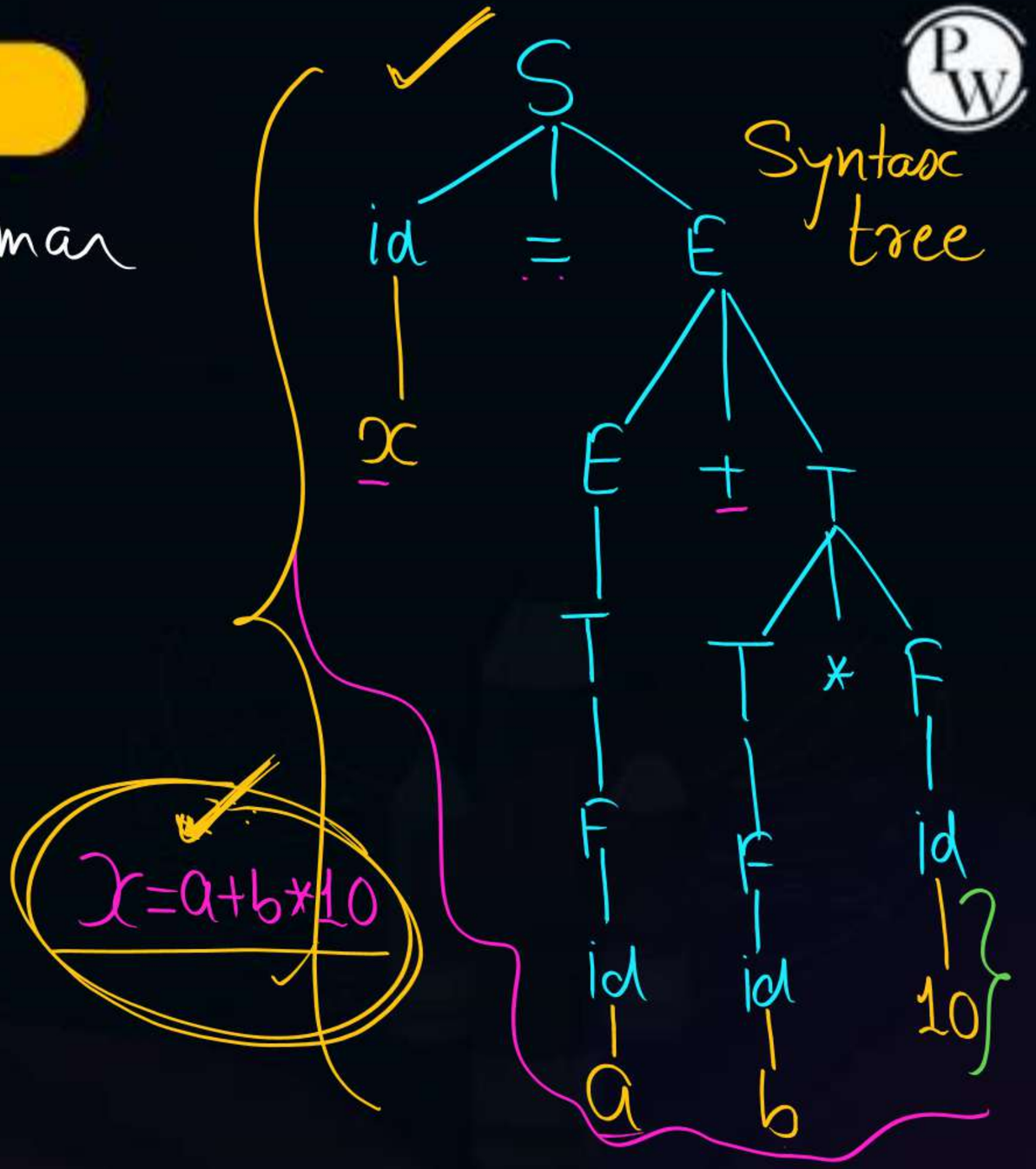
Grammar

$T \longrightarrow T*F/F$

$F \longrightarrow id$

and following input string

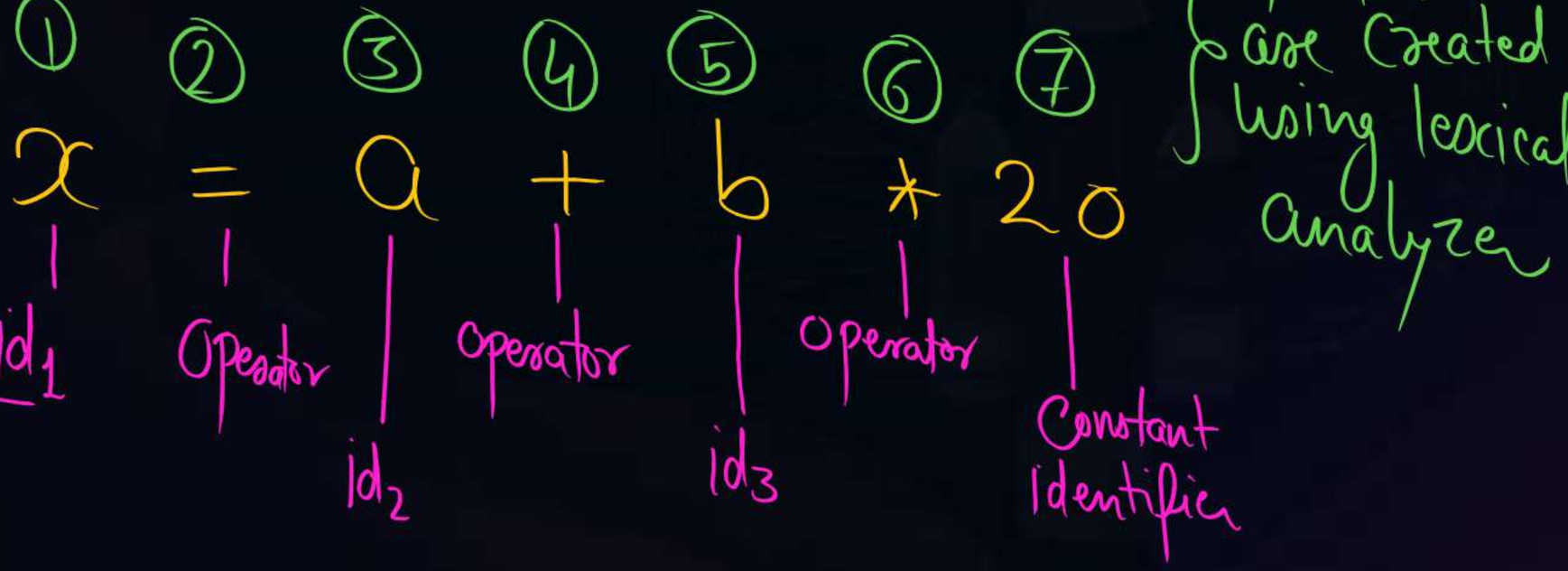
$x=a+b*10$



Slide

$x = a + b * 10$

(i) First of all this input string will be scanned by lexical analyzed and will be divided into tokens



Slide

(ii) Above stream of tokens will be given as an input to the syntax analyzer.

{Grammar is already known to the syntax analyzer}

* Syntax analyzer will create the parse tree for the input string based on the grammar rules.

{Same parse tree as previous slide will be created.}

{Parse tree can be generated only if given input string is a part of language generated by that grammar.}

Slide



2 mins Summary

Topic Parse tree(Concrete Syntax tree) vs Syntax tree(Abstract Syntax Tree)

Topic Objective of symbol table and error handles

Topic Process of translation

Topic Basic understanding of lexical analyzer

Topic

THANK - YOU